RoboPIN

Grounded Embodied Reasoning via Pinned Chain-of-Thought

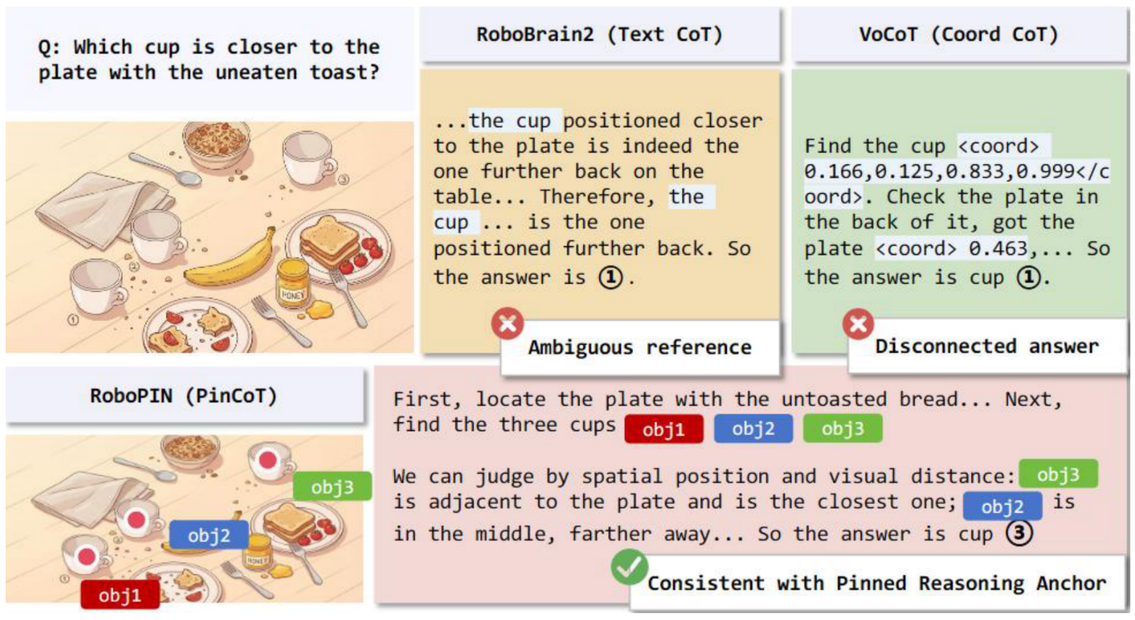
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Motivation



Paper Figure 1 · comparing three CoT paradigms

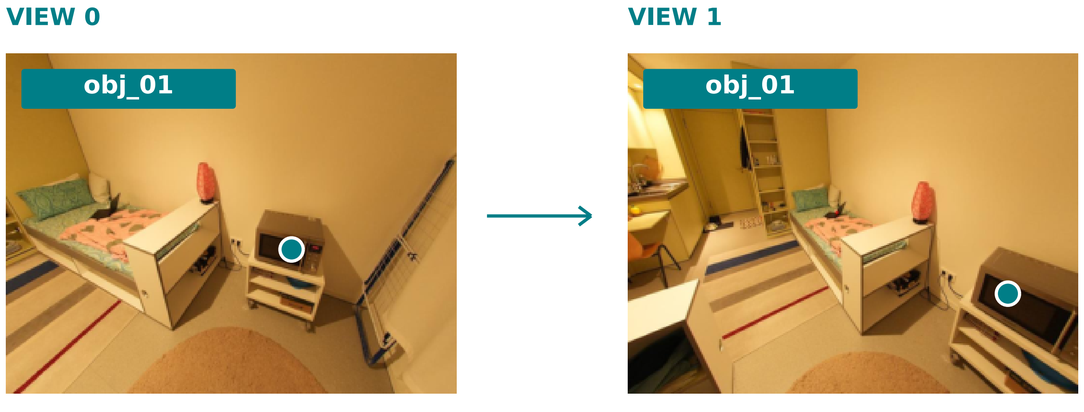
1. Reference drift

Textual references can switch between similar objects during multi-step reasoning.

2. Reasoning–answer misalignment

A correct intermediate analysis can still end on the wrong object, especially across views.

Key Insight: Persistent Identity



Same entity. Same ID. Different views.

Reasoning anchor = name + ID + view + point.

Introduce the anchor once; reuse its identity throughout the reasoning chain.

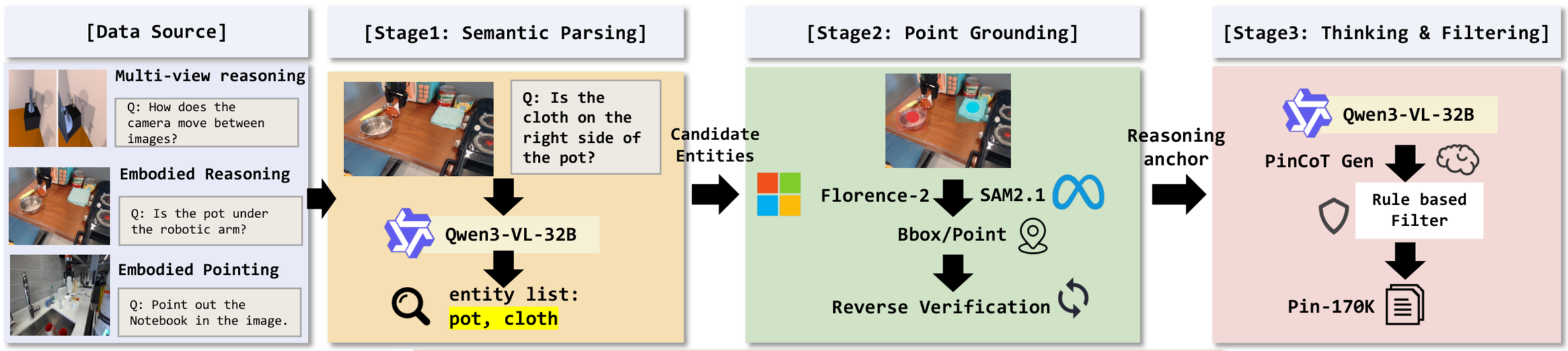
Our insight

Persistent IDs connect visual grounding, intermediate steps and the final answer.

QR PLACEHOLDER

PAPER / CODE

Our Method: RoboPIN / PinCoT

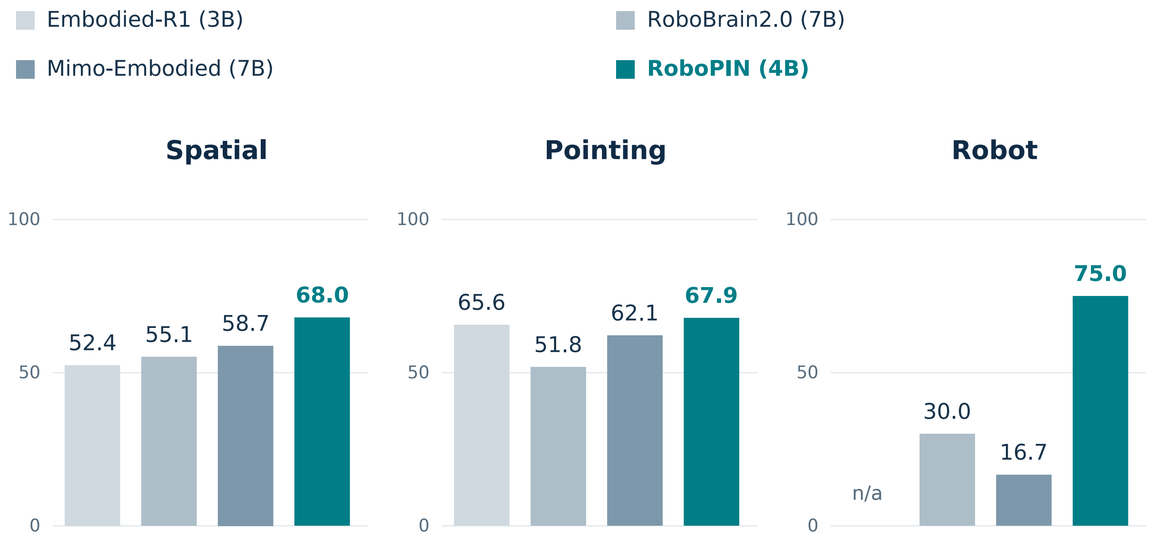


Three-stage post-training

SFT	→	CoT-SFT	→	RFT / GRPO
Embodied knowledge		Structured reasoning		Process alignment

PinCoT binds visual evidence to persistent identities. PIN-170K supplies structured traces; process rewards supervise grounding and identity consistency.

Experimental Results

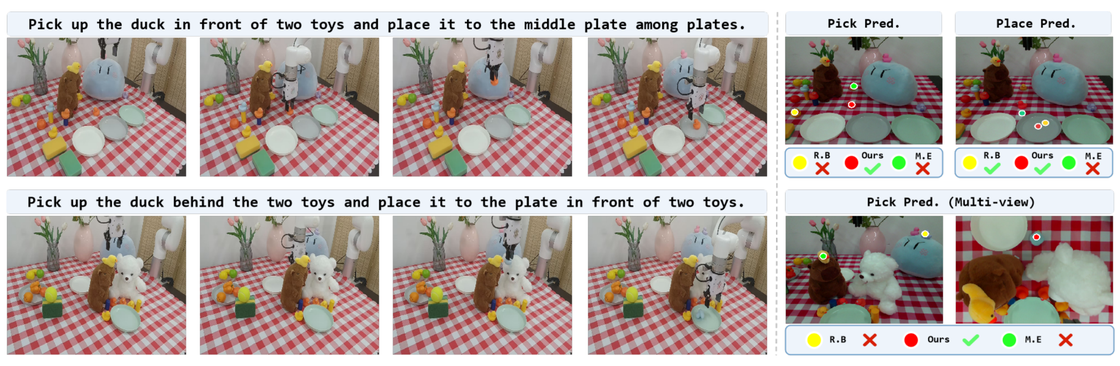


Scores (%) · Figure 1 / Tables 1–3. Spatial: 13-column mean; pointing: 5-benchmark mean; robot: success over 60 trials.

18.8% → 99.9% cross-step ID coverage

Without RFT → RoboPIN · EmbSpatial, Table 7

Qualitative Results



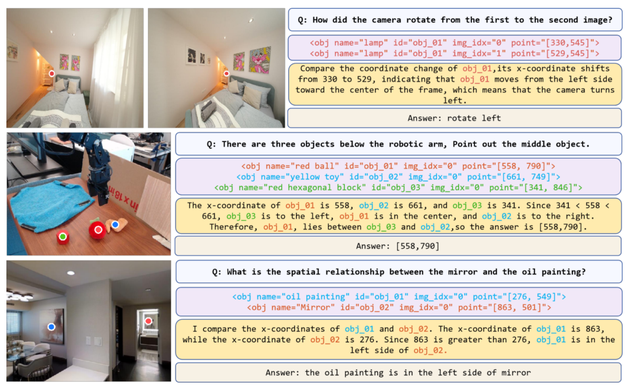


Figure 3: execution · Figure 4: grounded reasoning

Zero-shot xArm

Spatial
20 / 20

Multi-view
14 / 20

Long-horizon
11 / 20

Successes per task
Table 3

Persistent visual anchors make embodied reasoning traceable.